الأكاديمية السعودية الرقمية SAUDI DIGITAL ACADEMY | وزارة الاتصالات وتقنية المعلومات MINISTRY OF COMMUNICATIONS AND INFORMATION TECHNOLOGY | مهارات المستقبل | المؤسسة العامة للتدريب التقني والمهني Technical and Vocational Training Corporation | المركز الوطني للشراكات الاستراتيجية | WeCloudData

**شهادة إتمام تدريب**

# CERTIFICATE OF COMPLETION

**تبارك الأكاديمية السعودية الرقمية لـ**

# نوره نوار الهذلي

# Noura Nawar Alhuthali

إتمام **معسكر تدريبي في هندسة الذكاء االصطناعي** بنجاح بعد رحلة استمرت لمدة **7 أسابيع** في الفترة من **23 ذو القعدة 1447** حتى **10 محرم 1448**

On successfully completing **AI Engineer Bootcamp** in a journey that lasted **7 weeks** From **10 May 2026** to **25 June 2026**

تم إصدارها بتاريخ **10 محرم 1448**

Issue date **25 June 2026**

**WISHING YOU A CONTINUOUS SUCCESS**

**مع تمنياتنا بدوام التوفيق والنجاح**

**Amal A. Aljuhaiman**

الرئيس التنفيذي للأكاديمية السعودية الرقمية

امل الجهيمان



# Bootcamp Plan

## WEEK 1

### Pre-bootcamp Preparation

- Bootcamp orientation
- Software setup
- Self-paced e-learning of Python and ML fundamentals

## WEEK 2

### AI Engineering and Machine Learning Foundations

- AI Engineer role and AI lifecycle
- Introduction to machine learning
- Regression, classification, and model evaluation
- pandas, EDA, and feature engineering
- Hyperparameter tuning

## WEEK 3

### Model Experimentation and ML Workflows

- Decision trees, random forests, and ensemble methods
- Unsupervised clustering and K-means
- MLflow and experiment tracking
- Model selection, diagnostics, and iteration

## WEEK 4

### Deep Learning Fundamentals

- Neural networks and PyTorch basics
- Tensors, gradients, and model training workflow
- CNNs and model packaging
- Sequence models, RNNs, and LSTMs

## WEEK 5

### Generative AI, Transformers, Prompt Engineering, and RAG

- GenAI, LLMs, and AI ethics
- Transformers, BERT, and Hugging Face
- Prompt engineering and DSPy
- Embeddings, vector databases, and FAISS
- RAG and RAG evaluation

## WEEK 6

### LLM Applications, AI Agents, and AI App Prototyping

- LangChain and LangChain RAG workflows
- LLM evaluation
- Fine-tuning introduction, PEFT, and LoRA
- AI agents, LangGraph, and tool usage
- FastAPI and Streamlit

## WEEK 7

### Capstone Sprint and Final Presentation

- Architecture review and sprint kickoff
- Project implementation and solution integration
- Demo preparation and final presentation
- Course wrap-up